BayesINE9. Aplicaciones

Intro a Estadística Bayesiana

DataLab

Abril 2026

ICMAT-CSIC

Contents

- Chapter objectives
- Some applications from the BayOff report
- Prelims. CRC screening
- Bayesian networks for CRC risk mapping
- Influence diagrams for CRC screening
- Multiagent influence diagrams to incentivise CRC screening
- Final comments
- References

Chapter objectives

- Briefly mention some applications from the BayOff report
- Analysis in detail a current application

Some applications from the BayOff report

- Sampling
- Measurement and improvement of data quality
- Knowledge extraction
- Data disclosure
- Imputation
- Small area estimation
- Spatial statistics
- Unemployment forecasting
- Industrial accident forecasting
- Mortality forecasting
- Institutional evaluation
- Election forecasting
- Multisource data integration
- Visualisation
- Typology identification

Prelims. CRC screening

The issue. Colorectal cancer (CRC)

- Third most common cancer

- 2020. 1.9M new cases and 930000 associated deaths globally
- 2015. EU estimated annual costs 19B€
- Early-onset rising

- Early detection entails much less costs and deaths
- Only 14% susceptible EU citizens participate in screening programs

- Current scheme. Invite people 50-70. FIT; if positive, colonoscopy

The issue. Colorectal cancer (CRC)

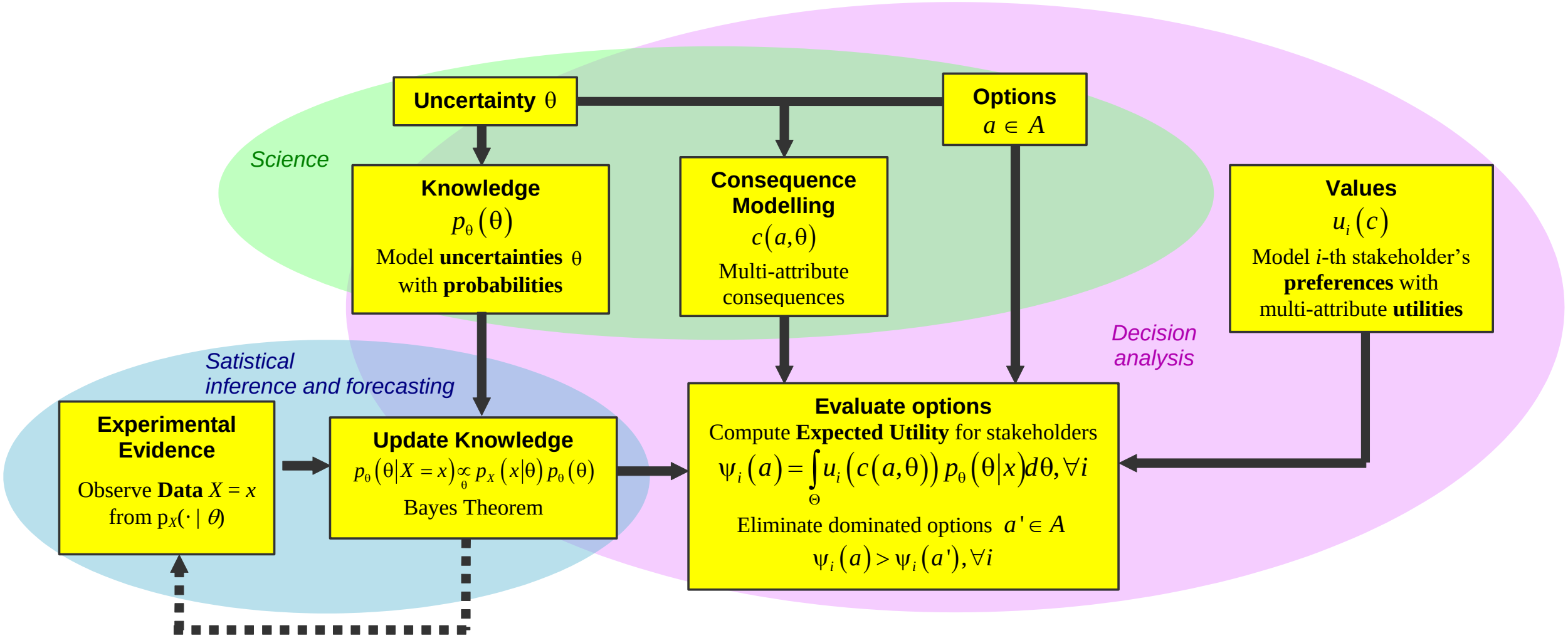
ONCOSCREEN

- New screening devices: Accurate, non-invasive, cost effective
- Raise further awareness

- More personalised screening
- Better screening resource allocation
- Incentives to nudge adherence to screening

Dataset

- Quirónsalud. 2012-2016 Annual health reviews
- Enriched with INE data postcode-based (Socioecon, Educational) Earlier
- Cleaned and consolidated (misrecorded, outliers, duplicates, missing values,…) Earlier
- 2M+ records. 66 variables (2012-2015 for model building, 2016 for testing)


Uncertainty θ
Options
$a \in A$
Science
Knowledge
$p_\theta(\theta)$
Model **uncertainties** θ
with **probabilities**
Consequence
Modelling
$c(a,\theta)$
Multi-attribute
consequences
Values
$u_i(c)$
Model *i*-th stakeholder's
preferences with
multi-attribute **utilities**
Decision
analysis
Satistical
inference and forecasting
Experimental
Evidence
Observe **Data** $X = x$
from $p_X(\cdot \mid \theta)$
Update Knowledge
$p_\theta(\theta|X=x) \underset{\theta}{\propto} p_X(x|\theta)\, p_\theta(\theta)$
Bayes Theorem
Evaluate options
Compute **Expected Utility** for stakeholders
$\psi_i(a) = \int_\Theta u_i(c(a,\theta))\, p_\theta(\theta|x)\, d\theta, \forall i$
Eliminate dominated options $a' \in A$
$\psi_i(a) > \psi_i(a'), \forall i$

Bayesian networks for CRC risk mapping

Probabilistic diagrams

- As basic tools for qualitative modelling of uncertainty use probabilistic influence diagrams a.k.a. causal networks, Bayesian networks, Belief networks,…. See the excellent

http://en.wikipedia.org/wiki/Bayesian_network

They are **influence diagrams** with chance nodes only. Qualitatively they describe a probabilistic model through

$$P(A1, A2,\ldots, An) = P(A1 \mid ant(A1))\ldots P(An \mid ant(An))$$

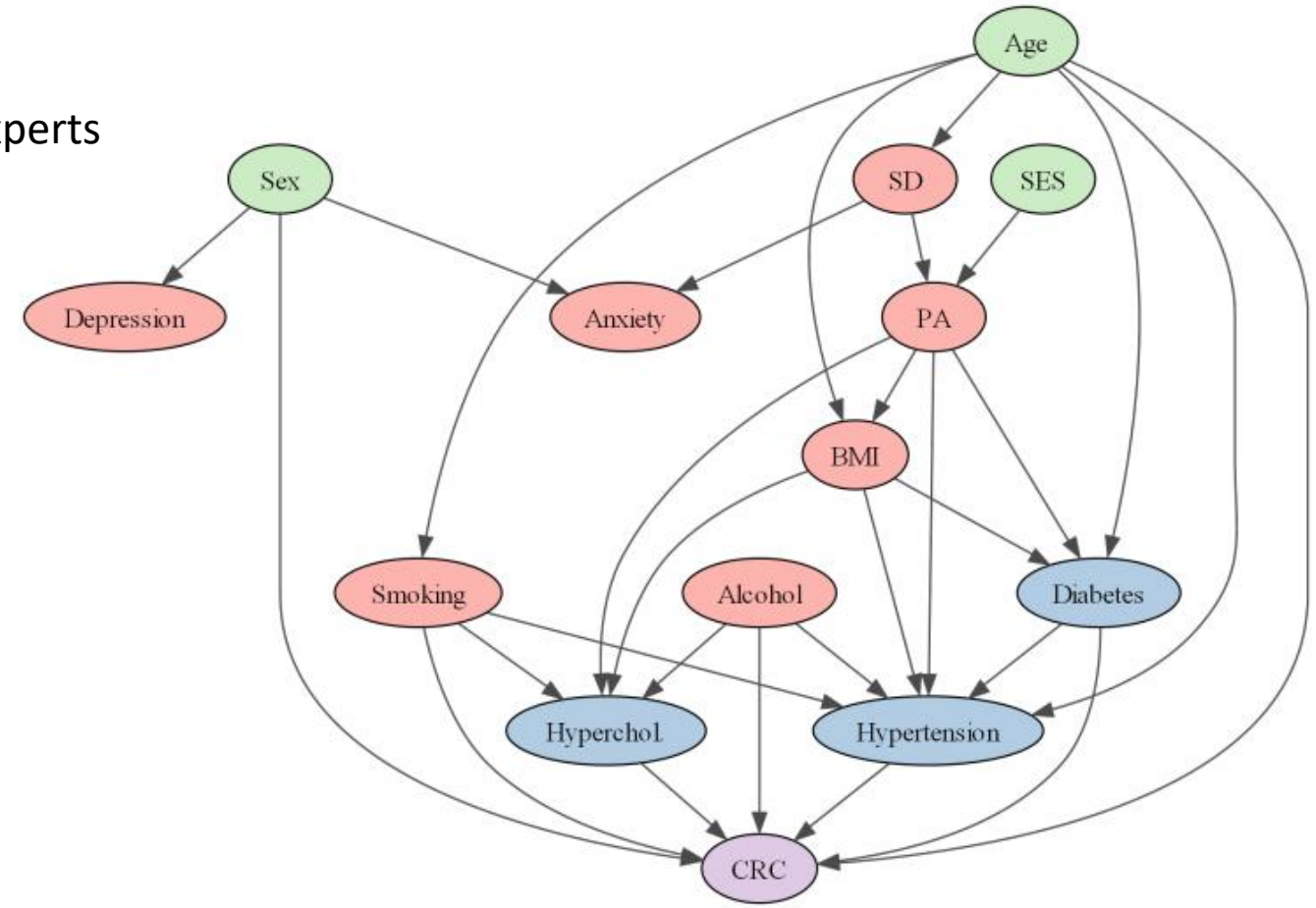
where ant (Ai) are the antecessors of node Ai.

Process

- Through bibsearch of risk factors, initial Bayesian network for CRC risk prediction
- Validated and corrected independently by several doctors, consensus sesión
- Initial solution for several structure extraction algorithms/tools enriched with our database 2012-2015
- Validated by several doctors
- Probability tables based on Dirichlet-Multinomial models
- Validated through classification on 2016 data
- Uses: Risk Maps and Influencial variables (as illustration)

Through standard bibsearch (adapted to database)

Then reviewed by experts

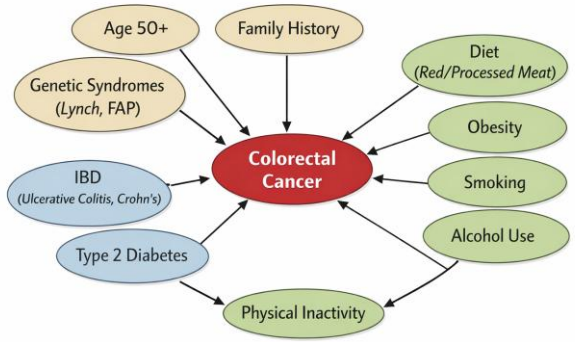


Non-modifiable
Modifiable
Medical conditions
CRC

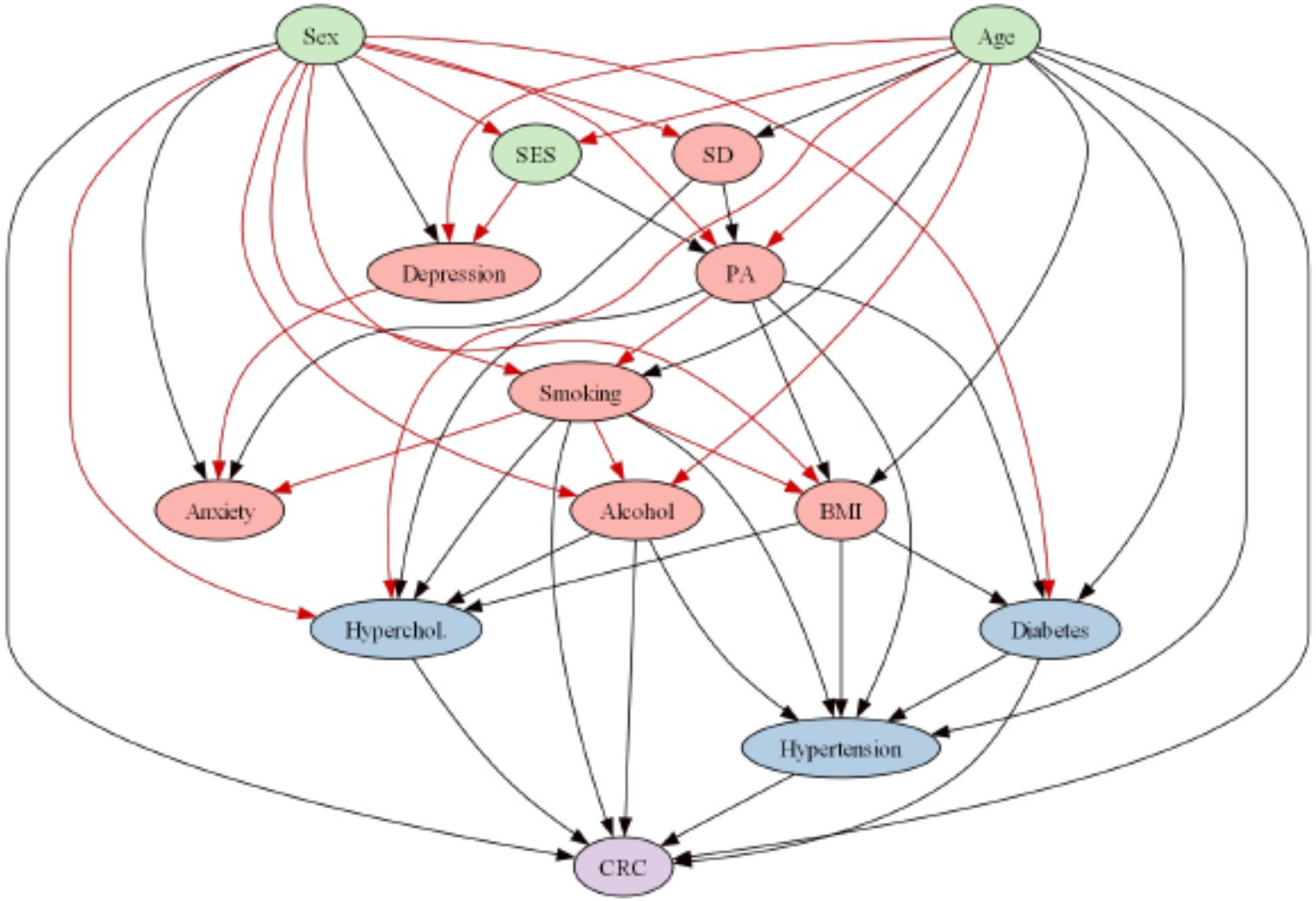
From ChatGPT….

- Colorectal cancer risk comes from a mix of **age, genetics, lifestyle, and certain medical conditions**. Here are the big ones, grouped so they're easier to digest:
-

Age & Genetics
Age 50+ (risk rises steadily with age, though rates are increasing in younger adults)
Family history of colorectal cancer or advanced polyps
Inherited syndromes, like:
Lynch syndrome
Familial adenomatous polyposis (FAP)
Personal Medical History
Previous colorectal cancer or adenomatous polyps
Inflammatory bowel disease (ulcerative colitis or Crohn's disease involving the colon)
Type 2 diabetes
Lifestyle Factors
Diets high in red or processed meats
Low fiber, fruit, and vegetable intake
Physical inactivity
Overweight or obesity
Smoking
Heavy alcohol use
Other Factors
Male sex (slightly higher risk than females)
African American and Ashkenazi Jewish ancestry (higher incidence and mortality, for complex genetic and social reasons)



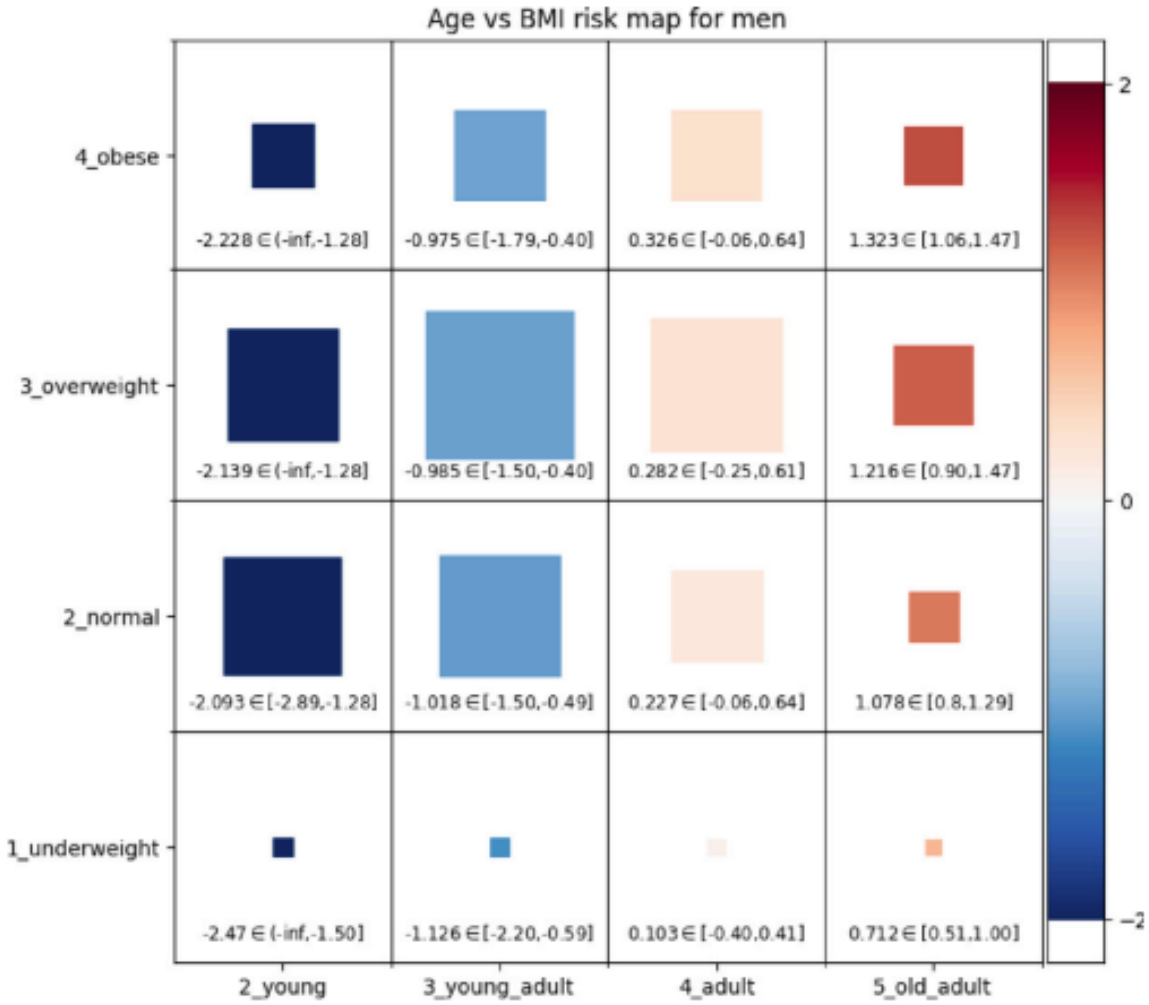
$$
\begin{aligned}
p(v_{sex},\ldots,v_{depression}) = {} & \Big[p(v_{sex})p(v_{age})p(v_{SES}|v_{sex},v_{age})\Big]\times \\
& \Big[p(v_{SD}|v_{sex},v_{age})p(v_{PA}|v_{sex},v_{age},v_{SD},v_{SES})p(v_{depr}|v_{sex},v_{age},v_{SES}) \\
& p(v_{smok}|v_{sex},v_{age},v_{PA})p(v_{alc}|v_{sex},v_{age},v_{smok}) \\
& p(v_{BMI}|v_{sex},v_{age},v_{PA},v_{smok})p(v_{anx}|v_{sex},v_{SD},v_{smok},v_{depr})\Big]\times \\
& \Big[p(v_{hypchol}|v_{sex},v_{age},v_{PA},v_{smok},v_{BMI},v_{alc})p(v_{diab}|v_{sex},v_{age},v_{PA},v_{BMI}) \\
& p(v_{hypten}|v_{age},v_{PA},v_{smok},v_{BMI},v_{alc},v_{diab})\Big]\times \\
& p(v_{CRC}|v_{sex},v_{age},v_{alc},v_{smok},v_{hypchol},v_{hypten},v_{diab}) \qquad (1)
\end{aligned}
$$



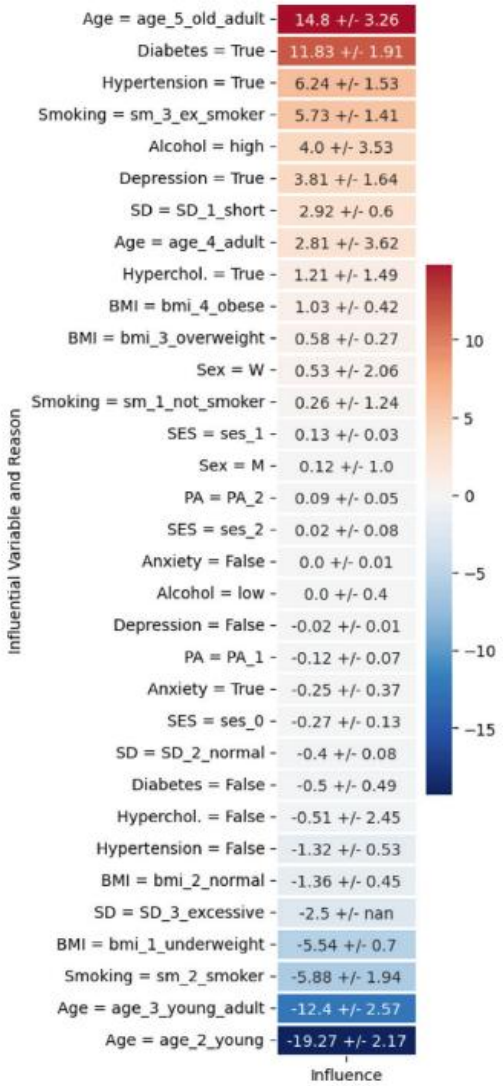
Probability tables
Priors

Validation

Risk maps


Age vs BMI risk map for men
4_obese
3_overweight
2_normal
1_underweight
-2.228 ∈ (-inf,-1.28]
-0.975 ∈ [-1.79,-0.40]
0.326 ∈ [-0.06,0.64]
1.323 ∈ [1.06,1.47]
-2.139 ∈ (-inf,-1.28]
-0.985 ∈ [-1.50,-0.40]
0.282 ∈ [-0.25,0.61]
1.216 ∈ [0.90,1.47]
-2.093 ∈ [-2.89,-1.28]
-1.018 ∈ [-1.50,-0.49]
0.227 ∈ [-0.06,0.64]
1.078 ∈ [0.8,1.29]
-2.47 ∈ (-inf,-1.50]
-1.126 ∈ [-2.20,-0.59]
0.103 ∈ [-0.40,0.41]
0.712 ∈ [0.51,1.00]
2_young
3_young_adult
4_adult
5_old_adult
2
0
−2

Influencial variables



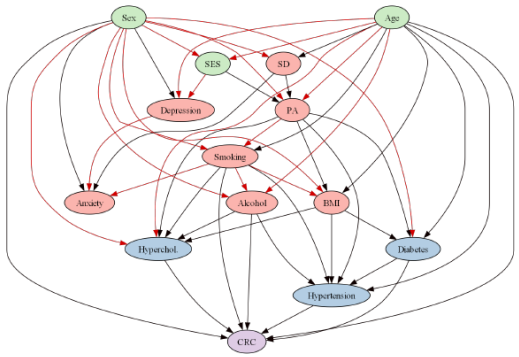
Influence diagrams for CRC screening

Influence Diagrams

- Tool to structure (and solve) decision making problems
- Direct acyclic graph G=(N,A)
- Three main types of nodes.
  - Chance. Circle
  - Decision. Square
  - Value. Hexagon, Diamond
    - Fourth type of node. Deterministic. Double circle
- Two types of arcs
  - Arcs into decision nodes
  - Arcs into chance and value nodes

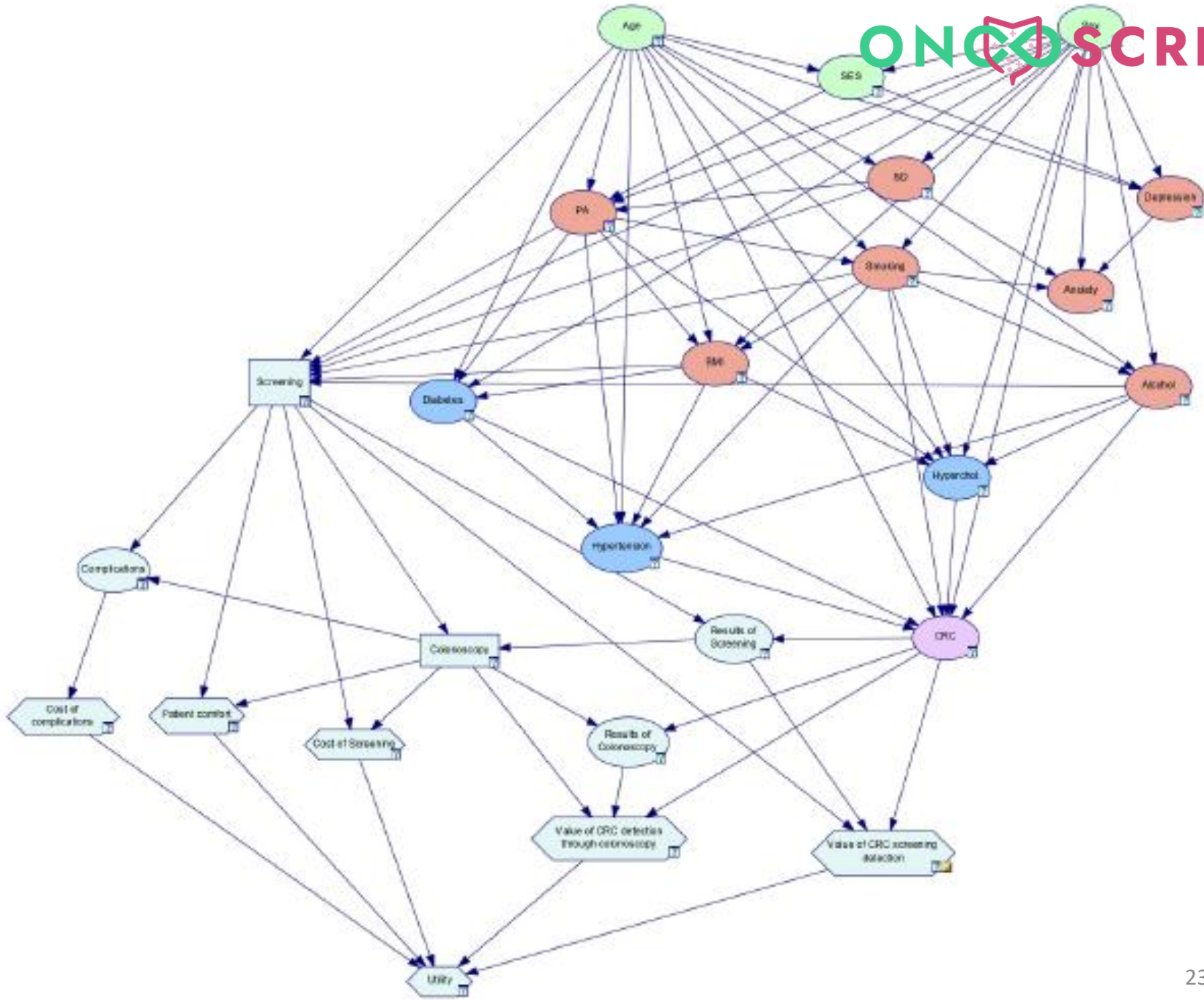
Process

- Construct the influence diagram
- Assess new probability tables (based on public sources)
- Identify objectives
- Construct utility function
  - Value function (for each comfort level)
  - Risk aversion
- Validate through sensitivity analysis
- Use cases
  - Individual cases
  - Validate current screening program
  - Design new screening program
  - Benchmarking new devices



Cost intervention
Cost complications
Comfort
Information

Multicriteria value function
Risk averse utility function

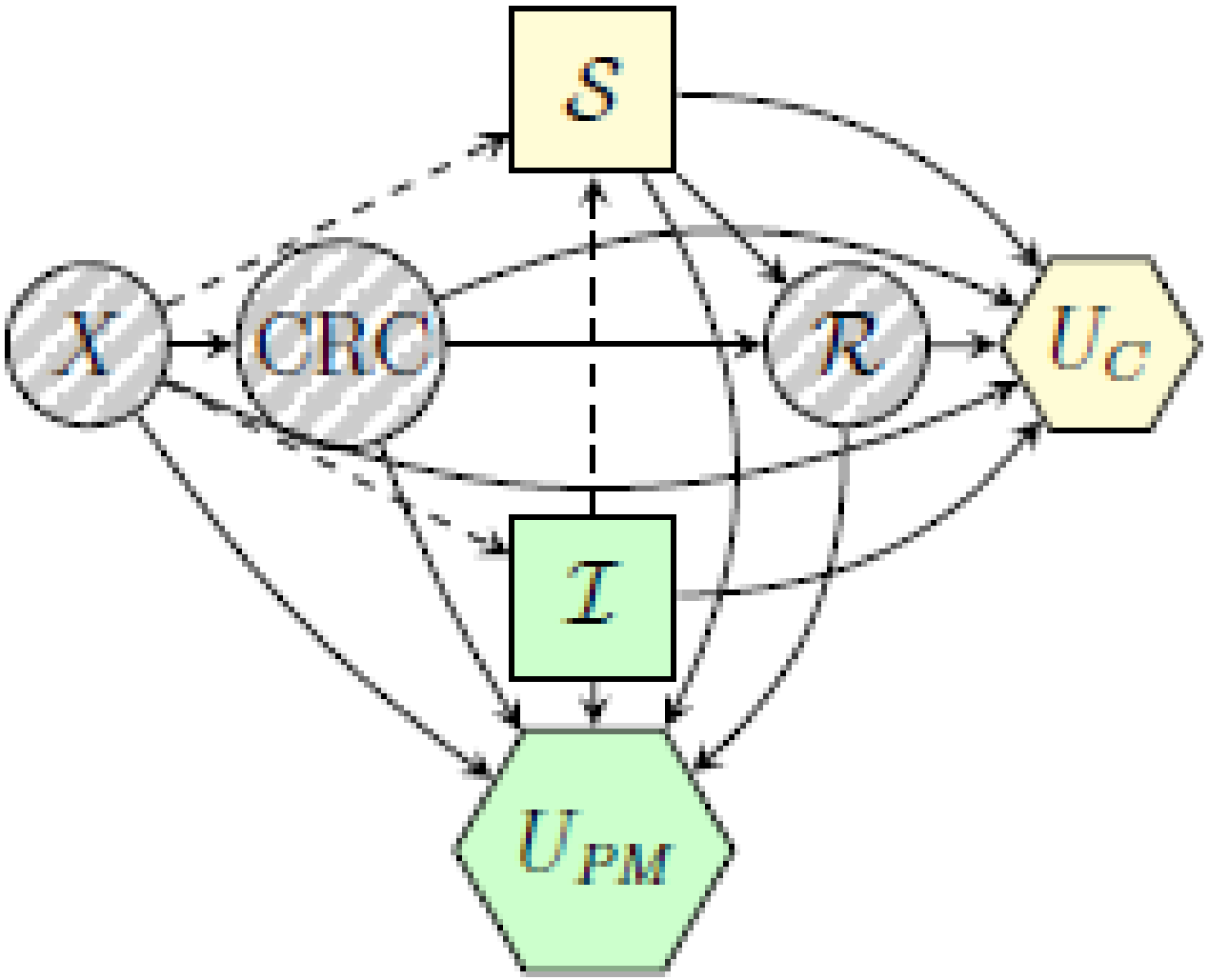


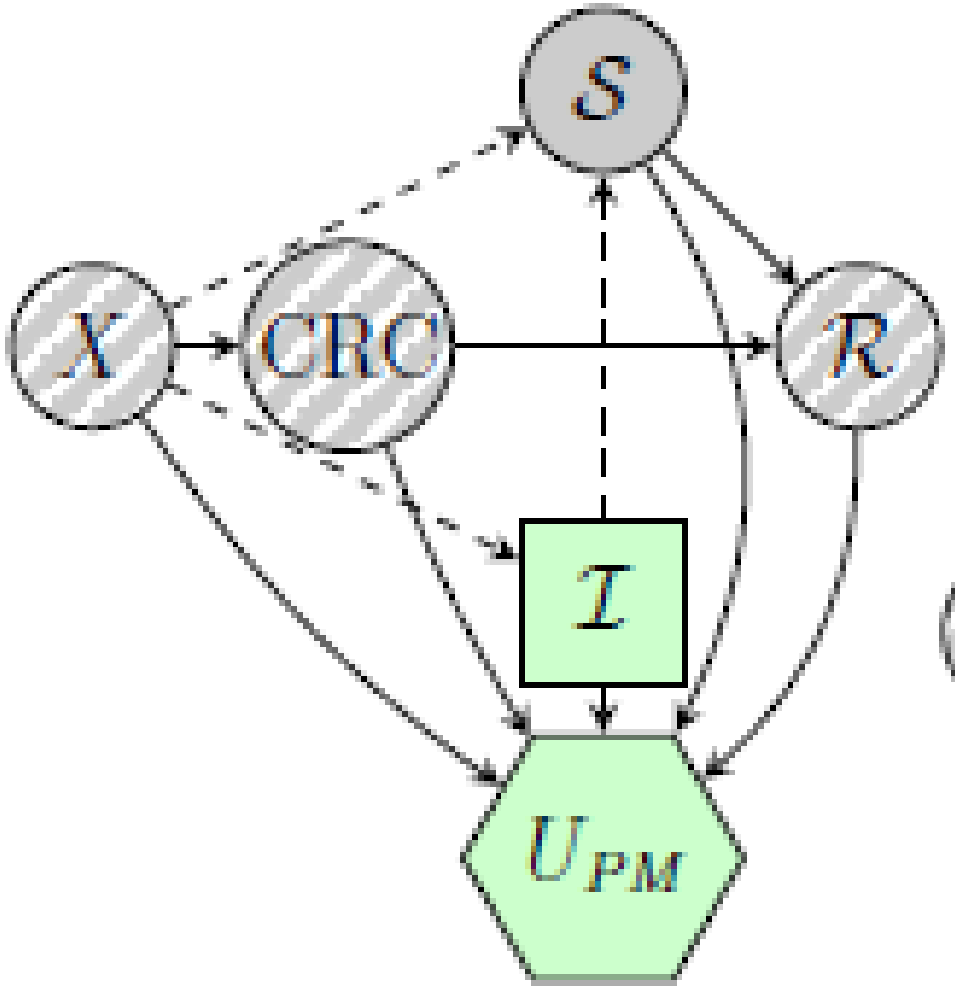
Use cases

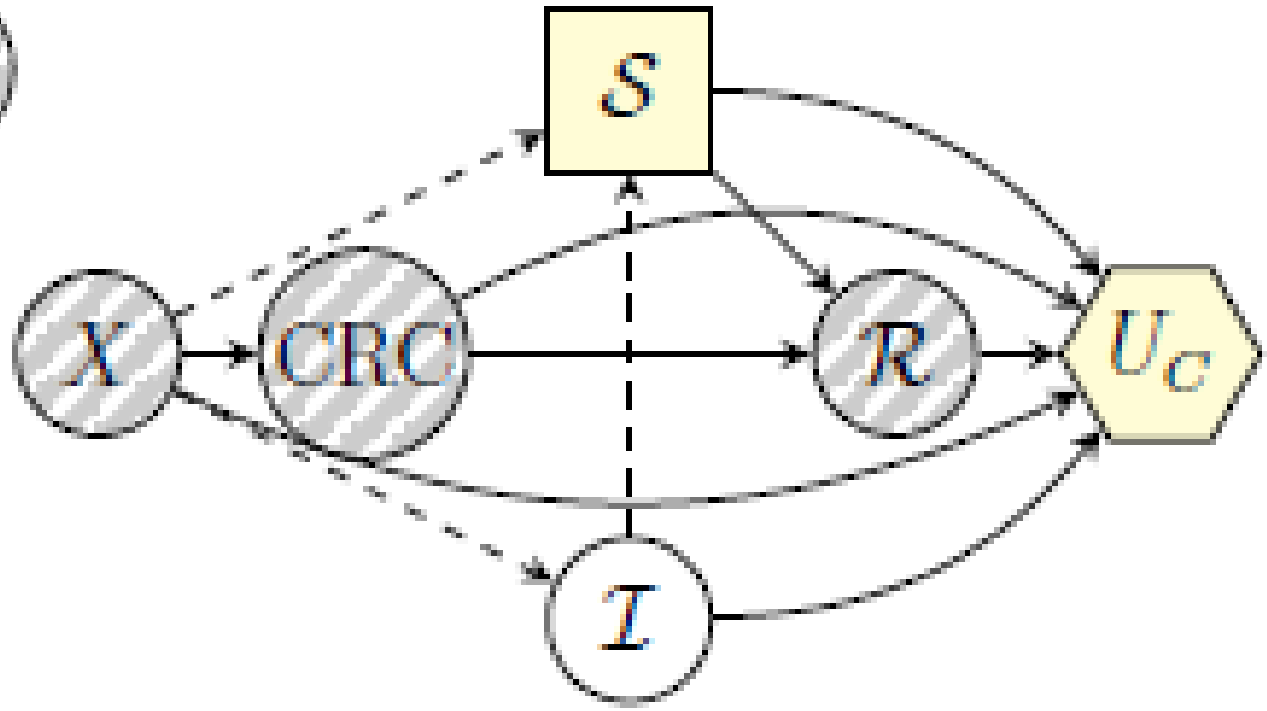
- Individual recommendations. Personalised screening based on risk
- Benchmarking new screening devices (ONCOSCREEN)
- Assessing screening programs (and designing)

Current Spanish Strategy	**Proposed Strategy**
• If patient >50 years old: Send FIT invitation regardless their features. • Else: No screening.	For patient with features X: • if $p(CRC\|X) < \theta_1 \rightarrow$ No screening • if $\theta_1 \leq p(CRC\|X) < \theta_2 \rightarrow$ FIT • if $p(CRC\|X) \geq \theta_2 \rightarrow$ sDNA
Example: Man under 50 with high risk (due to e.g. high alcohol consumption, overweight, exsmoking) would be detected by the model as a higher risk patient than a healthy man above 60.	
Results: Extrapolating results in Spain's target population, this strategy would detect 134 patients more.	

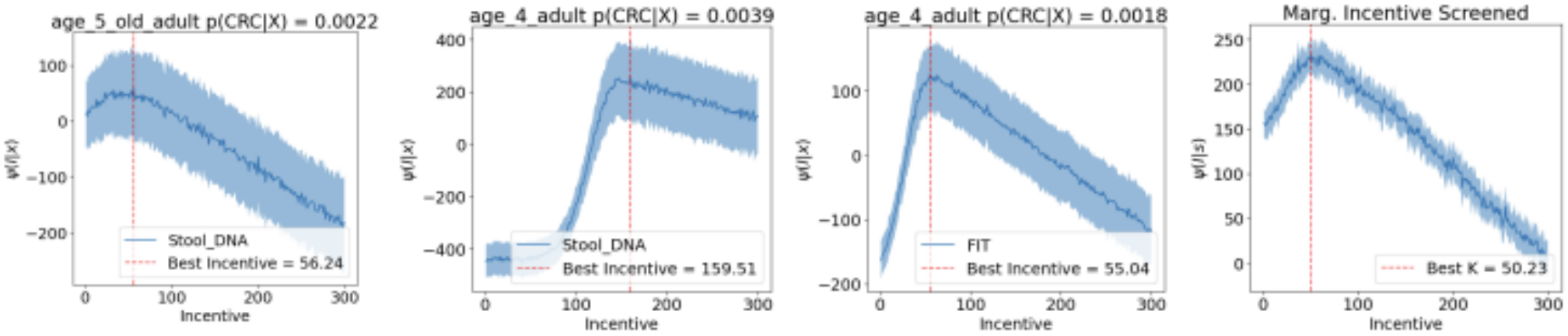
Multiagent influence diagrams to incentivise CRC screening


$\mathcal{S}$
$\mathcal{X}$
CRC
$\mathcal{R}$
U_C
$\mathcal{I}$
U_{PM}

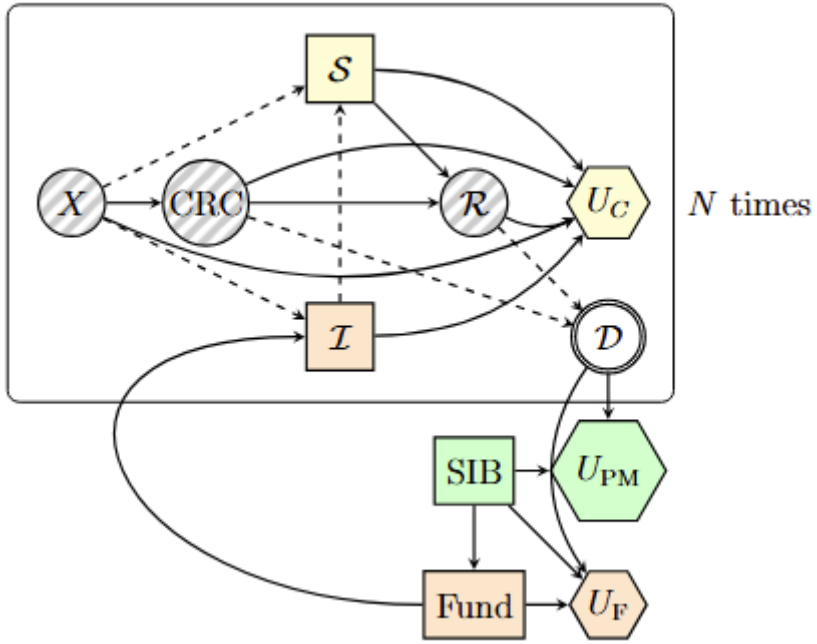

$\mathcal{S}$
$\mathcal{X}$
CRC
$\mathcal{R}$
$\mathcal{I}$
U_{PM}


$\mathcal{S}$
$\mathcal{X}$
CRC
$\mathcal{R}$
U_C
$\mathcal{I}$

Incentives


age_5_old_adult p(CRC|X) = 0.0022
Stool_DNA
Best Incentive = 56.24
Incentive
age_4_adult p(CRC|X) = 0.0039
Stool_DNA
Best Incentive = 159.51
Incentive
age_4_adult p(CRC|X) = 0.0018
FIT
Best Incentive = 55.04
Incentive
Marg. Incentive Screened
Best K = 50.23
Incentive

Incentives. Social impact bond



Final comments

Bayesian networks
Influence diagrams
Multi-agent influence diagrams

A very complex case in CRC screening
AI should end in supporting/automating decisions

Benchmarking the new devices

References

Colorectal cancer risk mapping through Bayesian networksD Corrales, A Santos-Lozano, S López-Ortiz, A Lucia, DR Insua

Computer Methods and Programs in Biomedicine, 108407

A decision support model for colorectal cancer screeningD Corrales, DR Insua, MJ González

Computers in Biology and Medicine 196, 110755

Incentivising Personalised Colorectal Cancer Screening: an Adversarial Risk Analysis ApproachD Corrales, DR Insua

2025 IEEE 38th International Symposium on Computer-Based Medical Systems

Thanks so much

Should you have any questions

david.rios.insua@gmail.com
pablo.garcia@icmat.es
mario.chacon@icmat.es

We love helping (and messing) with complex problems!!!